

QUESTION-7

Single linked list using insertion in at any point

main.c	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3- struct node { 4 int data; 5 struct node *next; 6 }; 7 struct node *head = NULL; 8- void insertBegin(int value) { 9 struct node *newNode = (struct node 10 *)malloc(sizeof(struct node)); 11 newNode->data = value; 12 newNode->next = head; 13 head = newNode; 14 } 15- void insertEnd(int value) { 16 struct node *newNode = (struct node 17 *)malloc(sizeof(struct node)); 18 struct node *temp = head; 19 newNode->data = value; 20 newNode->next = NULL; 21- if (head == NULL) { 22 head = newNode; 23 return; 24 } 25 }</pre>	<pre>Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL === Code Execution Successful ===</pre>
<pre>23 while (temp->next != NULL) 24 temp = temp->next; 25 temp->next = newNode; 26 } 27- void insertPosition(int value, int pos) { 28 struct node *newNode, *temp; 29 int i; 30 newNode = (struct node *)malloc(sizeof 31 (struct node)); 32- newNode->data = value; 33- if (pos == 1) { 34 newNode->next = head; 35 head = newNode; 36 return; 37 } 38 temp = head; 39 for (i = 1; i < pos - 1 && temp != NULL; i 40 ++){ 41 temp = temp->next; 42 } 43- if (temp == NULL) { 44 printf("Invalid Position\n"); 45 return; 46 } 47 }</pre>	<pre>Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL === Code Execution Successful ===</pre>
<pre>57 return; 58 } 59 60 printf("Linked List: "); 61- while (temp != NULL) { 62 printf("%d -> ", temp->data); 63 temp = temp->next; 64 } 65 printf("NULL\n"); 66 } 67 68- int main() { 69 insertBegin(10); 70 insertBegin(5); 71 72 insertEnd(20); 73 insertEnd(30); 74 75 insertPosition(15, 3); 76 77 display(); 78 79 return 0; 80 }</pre>	<pre>Linked List: 5 -> 10 -> 15 -> 20 -> 30 -> NULL === Code Execution Successful ===</pre>